Averages of neighbours

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Let $n > 2$ be an integer. Consider the map T :

$$(x_1, \ldots, x_n) \mapsto \left(\frac{x_1 + x_2}{2}, \frac{x_2 + x_3}{2}, \ldots, \frac{x_n + x_1}{2} \right).$$

Show that for every $x = (x_1, \ldots, x_n)$ whose x_i are not all equal, there is an integer k such that $T^k(x)$ has a non-integer component.